$\{5, 2, 4, 9\}$

$\{a, x, b, d\}$

$f(5) = a$
$f(2) = x$
$f(4) = b$
$f(9) = d$

} one-to-one onto } correspondance

n	$f(n)$
$\mathbb{N}$	$\mathbb{N}_\varepsilon$
1	2
2	4
3	6
4	8
5	10
6	12
⋮	⋮

$f(n) = 2n$

$\mathbb{N}$	$\mathbb{N}_\varepsilon$
1	4
2	2
3	6
4	8
5	10
⋮	⋮

$\mathbb{N}$	$\mathbb{N}_\varepsilon$
1	8
2	4
3	10
4	6
5	2
6	12
7	⋮
⋮	

$\mathbb{N}$	$\mathbb{N}_\varepsilon$
1	4526
2	28
3	444
4	12345
5	7278
⋮	⋮

$\mathbb{N}_\mathbb{Z}$ is countable.

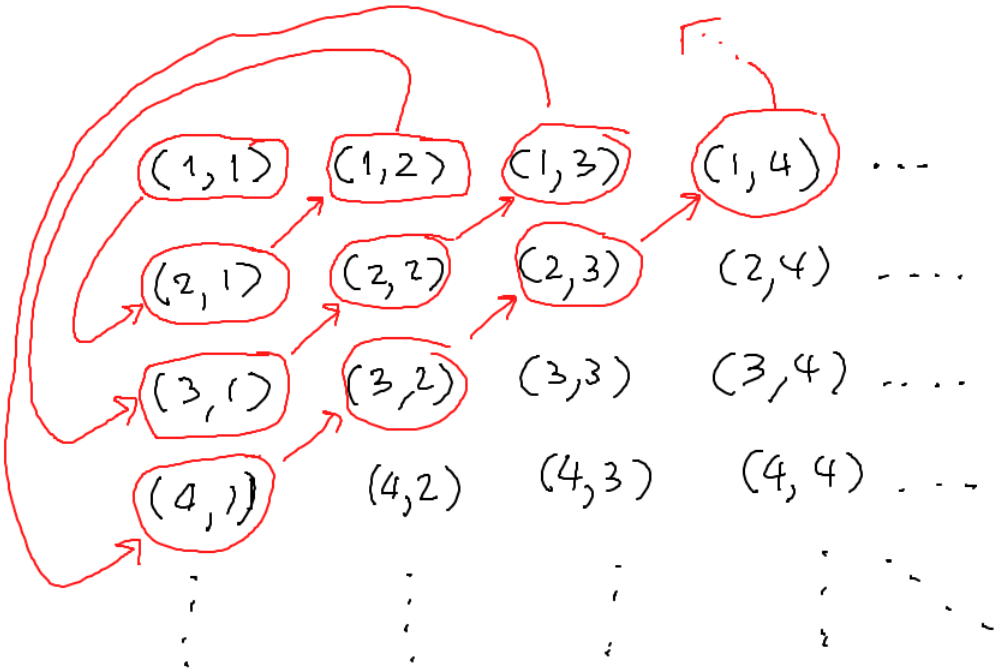
Show that $\mathbb{Z}$ is countable. Need to show that there exists a correspondence between $\mathbb{N}$ and $\mathbb{Z}$

$\mathbb{N}$	$\mathbb{Z}$
1	0
2	1
3	2
4	3
5	4
6	5
⋮	⋮
?	-1
	-2
	-3
	⋮

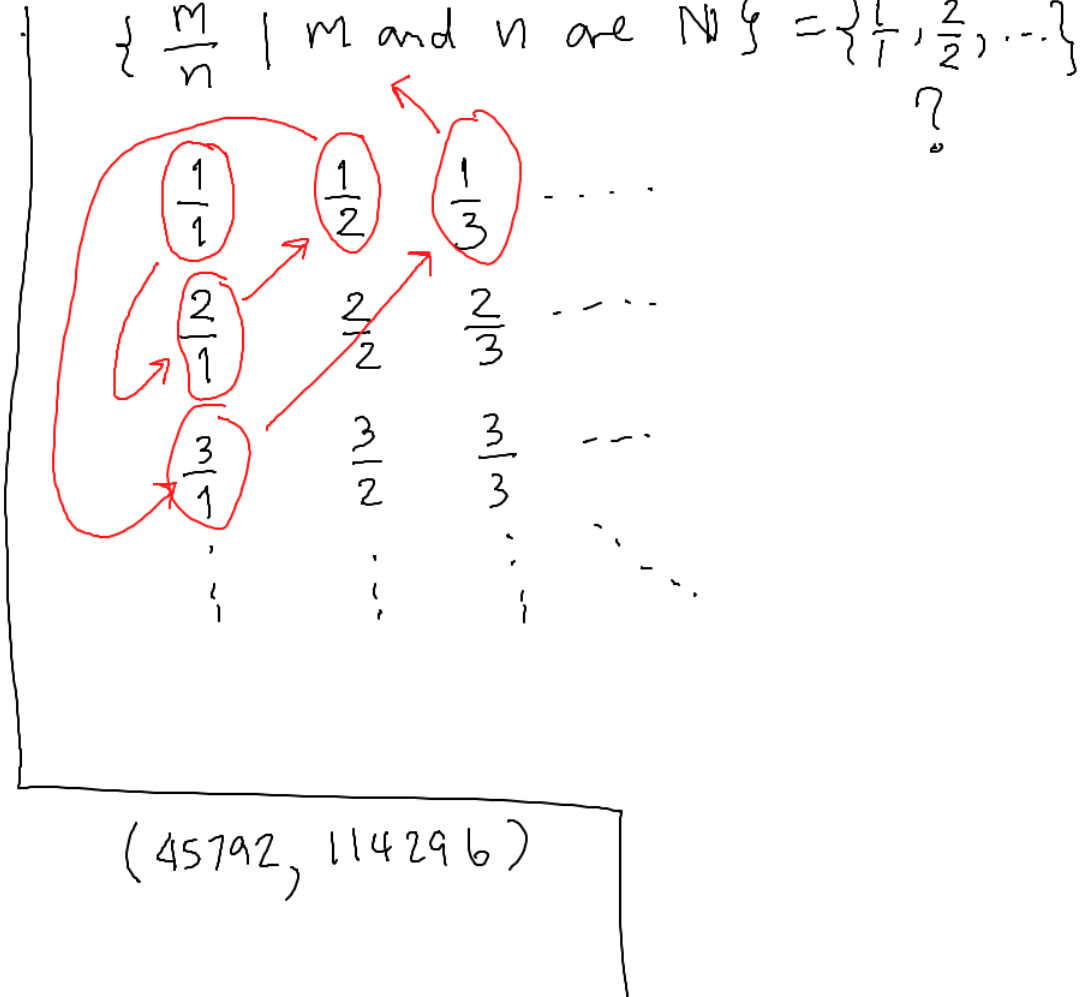
invalid correspondence

$\mathbb{N}$	$\mathbb{Z}$
1	0
2	1
3	-1
4	2
5	-2
6	3
7	-3
8	4
9	-4
? ←	-157964

$\mathbb{N} \times \mathbb{N} = \{(x, y) \mid x \in \mathbb{N} \text{ and } y \in \mathbb{N}\}$



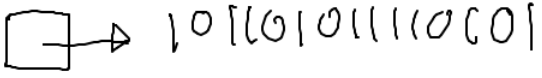
$\left\{\frac{m}{n} \mid m \text{ and } n \text{ are } \mathbb{N}\right\} = \left\{\frac{1}{1}, \frac{2}{2}, \ldots\right\}$?



$\Sigma = \{0, 1\}$

$\Sigma^* = \{\varepsilon, 0, 1, 00, 01, 10, 11, \ldots\}$

$\mathbb{N}$	Σ^*
1	ε
2	0
3	1
4	00
5	01
6	10
⋮	⋮



The set of all TMs

"$(Q, \Sigma, \Gamma, \delta, q_0, q_a, q_r)$"

IBS = the set of all infinite binary sequences

01101111011111...
0000000....
1111...—

$\mathbb{N}$	IBS
1	00000..
2	10000..
3	01000...
4	11000..
5	00100.—
6	10100...
7	01100...
⋮	

Assume that IBS is countable. Since IBS is countable there exists a correspondence between $\mathbb{N}$ and IBS.

$X = 11111... \in IBS$

x does not correspond with a natural # n because the nth bit (from left) of x is different than the nth bit (from left) of the infinite binary sequence that corresponds with n.

The set IBS is uncountable.

$\mathbb{R}$
123.0000000...
3.14159265394...
1.414...

The set of all languages

$\Sigma = \{0, 1\}$

$\Sigma^* = \{\epsilon, 0, 1, 00, 01, 10, 11, 000, 001, 010, 011, 100, 101, 110, 111, 0000, \ldots\}$

$0\Sigma^* = \{0, 00, 01, 000, 001, 010, 011, 0000, \ldots\}$

0 1 0 1 1 0 0 1 1 1 1 0 0 0 0 1 ...

$\Sigma^* = 1111\ldots$

$\emptyset = 0000\ldots$

$\{\varepsilon\} = 1000\ldots$